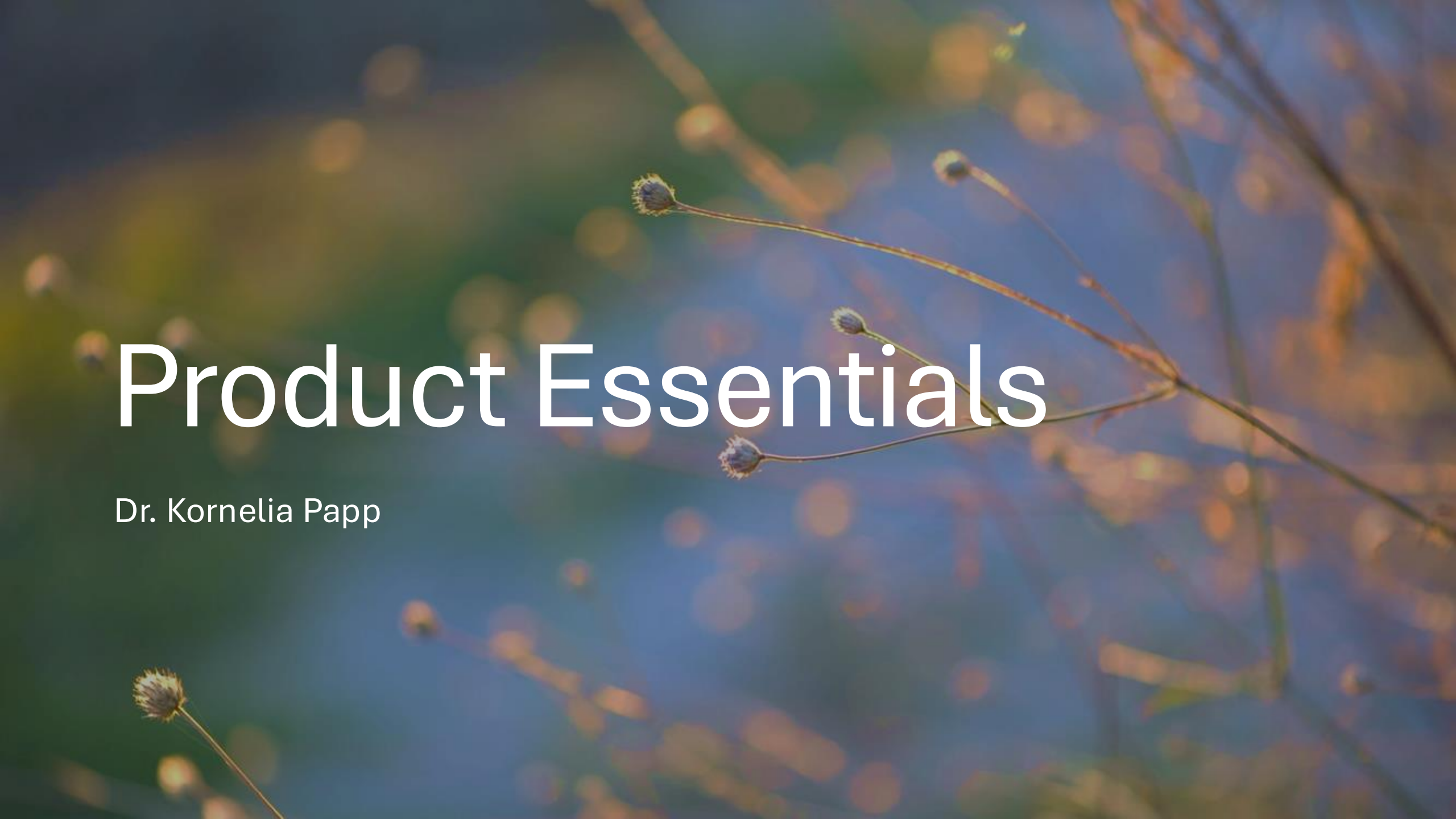
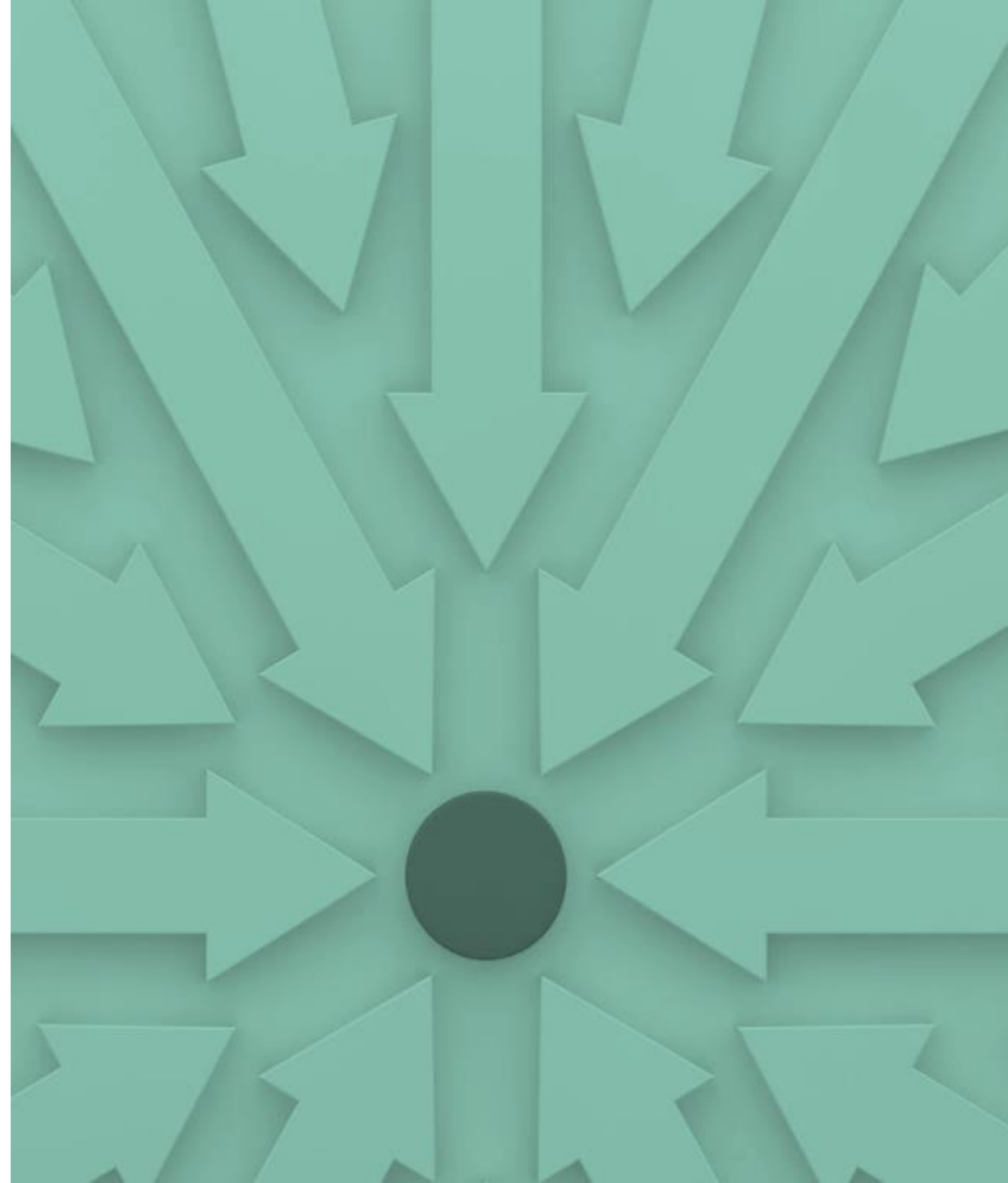




Product Essentials

Dr. Kornelia Papp

Building products
not (just) with AI
but with **purpose.**



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Session Goals

Understand how to move from an initial idea to a structured product opportunity by connecting **use case development**, **value assessment**, and **product development** into one coherent decision-making process.

What is being built?

Start with **use case development**: identifying the user, the core problem, the context, and the specific situation where AI or a product creates measurable value.

Should it be built?

Perform a **value assessment**: evaluating whether the use case is sufficiently important, feasible, and valuable to justify the investment of resources.

How is it built?

Move into **product development**: transforming the validated use case into a concrete concept by defining features, testing assumptions, and planning implementation.



Product development is rarely linear. Using AI to rapidly test and pivot is key.

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Use Case Development

Start with the
problem, not the
model.

User Need × AI Strength Framework

- Find good AI use cases before jumping to a tool
- A good AI use case is a real user problem × a task where AI has a structural advantage

“We have AI → where can we use it?” ❌

“We have a problem → does AI help?” ✅

- AI Strength: scale, speed, pattern
- Cost of failure: where does AI fail?
- Tolerance of error
- Human–AI complementarity
- **Don’t ask where to use AI. Ask where AI is structurally better than humans.**

Define User Need

What is the real user problem? Is an AI-based solution better than conventional software or process alternatives?

Question

Example

Who is the user?

e.g. student, doctor, customer-support agent

What is their painful task?

repetitive, uncertain, slow, cognitively heavy

What does success look like?

faster, cheaper, better, safer, more personalized

Why AI?

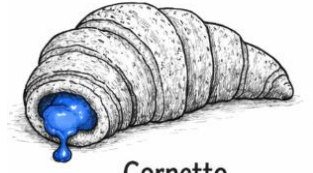
prediction, generation, classification, summarization, retrieval, automation

What should the human still control?

final decision, validation, exceptions



Croissant au Beurre
(French Classic)



Cornetto
(Italian Style)



Cronut



Cruffin



Flaffin



Gipfeli

Task Decomposition

A product is rarely a single item. Even a simple product is often a bundle of distinct components.

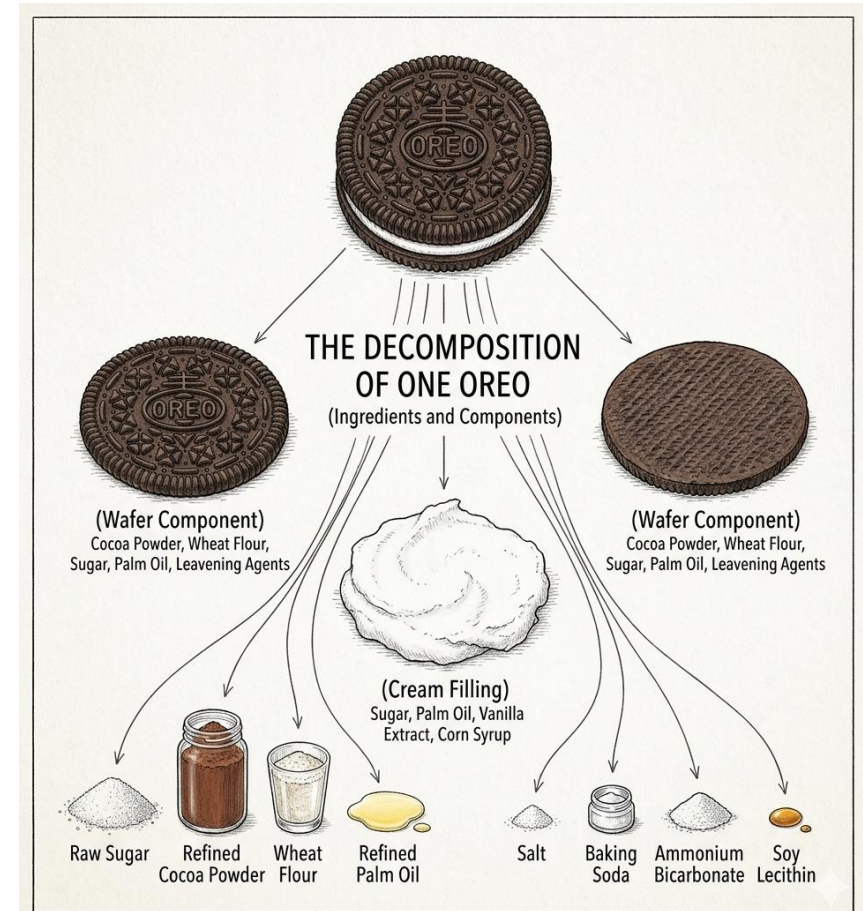
Forced decomposition: The purpose of decomposition is to find the *one* specific step where AI has a structural advantage.

Example: AI might be mediocre at "Writing a Roadmap," but it is world-class at "Extracting key points from 50 pages of notes."

Decomposition: Identify every discrete step required to produce the "Product."

Evaluation: Determine which sub-tasks are repetitive, high-volume, or data-heavy.

Integration: Apply AI only to the specific steps where it offers a structural advantage.



AI Capability Mapping

Indicators of a Strong AI Use Case

Target tasks where human constraints create a bottleneck:

- ✓ **High Frequency:** Tasks performed repeatedly where automation yields massive time savings.
- ✓ **Cognitive Drudgery:** Mentally taxing work that exhausts human focus but is trivial for a machine.
- ✓ **Probabilistic Logic:** Scenarios involving nuance and ambiguity that cannot be solved with rigid, "if-then" rules.
- ✓ **Massive Scale:** Data inputs or volumes that exceed human processing capacity.
- ✓ **Velocity Demands:** Situations requiring near-instant turnaround under significant time pressure.

Example: Report generation task

Task Step (Ingredients)	AI Fit / Structural Advantage
Reading large text	✓ Summarization
Extracting key points	✓ Pattern Recognition
Structuring output	✓ Generation
Verifying correctness	✗ Human-in-the-loop required

Human Role

Red Flags for a Weak AI Use Case

Avoid using AI when it adds unnecessary complexity or risk:

-  **Deterministic Tasks:** If a problem has a 100% predictable, rule-based solution, standard software is more efficient and reliable.
-  **Ungoverned High Stakes:** Safety-critical or high-liability decisions should never be fully automated without human oversight.
-  **Low Friction:** AI implemented for "novelty" that does not solve a documented user pain point.
-  **Vague ROI:** Projects lacking a clear, measurable benefit to the product's value or efficiency.

Example: Medical Diagnostic		
Phase	By	Explanation
Detection	AI	Handles massive input size and repetition that would cause human fatigue.
Prioritization	AI	Navigates ambiguity to surface only the most relevant or high-risk data points.
Verification	H	Ensures safety-critical decisions are governed by ethics and context, not just probability.
Decision	H	Manages scenarios where AI structural advantage fails or data is incomplete.

A Good AI Use Case

A good AI use case = A validated user pain point × A task where AI has a structural advantage.

1. **Identify the pain:** What is the specific bottleneck in the current "Product" (Code, Roadmap, Design, or Risk Model)?
2. **Verify the Advantage:** Is AI better here due to **scale**, **speed**, or **consistency**?
3. **Assess the Failure:** What happens if the AI is wrong? Is there a human-in-the-loop "Quality Gate"?
4. **Define the Complementarity:** Does using AI here free the human to focus on **strategy**, **empathy**, or **ethics**?

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Define Success

Quantifying Success

A use case without a metric is a weak use case. To move from a pilot to a product, success must be defined by measurable improvement.

Primary Metrics

Efficiency (Time Saved): Does the AI-assisted workflow reduce the end-to-end processing time for a specific task?

Accuracy (Error Reduction): Does the model effectively lower the rate of human error in high-volume processing?

Utility (Output Quality): Is the resulting product (e.g., a report or a diagnostic) more comprehensive or personalized than the manual alternative?



If success cannot be quantified, the use case is insufficient for development. Measurable benefits confirm structural advantage.

Value × Feasibility × Risk Prioritization Matrix

Which AI use case should you build first?

Framework	Best for	How to calculate value
1. ROI / NPV business case	Financial value	Estimate revenue gain, cost saving, risk reduction, then subtract build/run/change costs. Use NPV if benefits arrive over years.
2. RICE	Product / AI feature prioritization	Reach × Impact × Confidence / Effort.
3. WSJF	Agile / portfolio sequencing	Cost of Delay / Job Size. Prioritize high urgency, low effort use cases.
4. Desirability–Viability–Feasibility	Innovation screening	Score: user need, business value, technical feasibility. Good first filter before detailed ROI.
5. Real Options	Uncertain tech / AI / R&D use cases	Value small experiments that create future choices: learn, scale, stop, pivot. Useful when ROI is unknown early.

Pick boringly useful before impressively futuristic.

Score & Classify

Score every idea from 1–5

Dimension	Ask
Value	Would this save time, money, improve quality, or unlock something new?
Feasibility	Do we have the data, tools, skills, and integration path?
Risk	Could errors, privacy issues, bias, or over-automation cause harm?
Time to impact	Can we test it quickly?

Classify ideas

Type	Decision
High value, high feasibility, low risk	Start here
High value, low feasibility	Research/prototype
Low value, high feasibility	Avoid “toy AI”
High risk	Needs governance or human-in-the-loop



Evaluate & Discuss

- Form groups of four/five
- Review all proposals (on next slide)
- Rank proposals in Excel board:
https://1drv.ms/x/c/03fc64168f48aeed/IQAOYvZBrnXeSpJc5wsID_WvAXtnnRC25FBQkzzmnttQgKc

Proposals

Project A: The Legacy Data Cleaner. An AI that automatically reformats old, messy actuarial spreadsheets into standardized databases.

Project B: The "Vibe" Bot. A chatbot for the internal HR portal that answers "how are you feeling?" with inspirational quotes.

Project C: Automated Claims Adjuster. An AI that automatically approves or denies medical claims under \$500 without any human review.

Project D: Customer Churn Predictor. A model that uses historical behaviour to predict which high-value clients are likely to leave in the next 30 days.

Template

Section	Answer
User	
Painful task	
Current workaround	
Task breakdown	
AI strengths applied	
Human in the loop	
Expected value	

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AI-driven Product Development

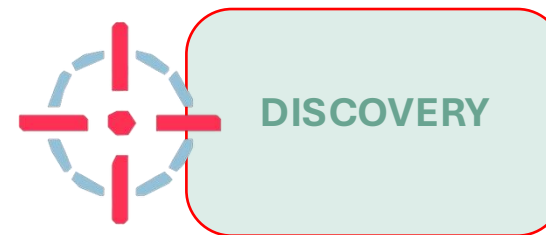
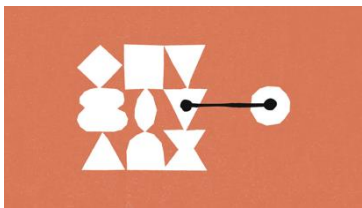
DISCOVERY

PROTOTYPE

ITERATE

EVALUATE

AI changes the process because outputs are probabilistic and quality depends on context, workflow, and trust.



Aspect		Traditional product development	AI-driven product development
 Process flow		Discovery → development → launch.	Discovery → prototype → evaluate → learn → iterate.
 Requirements		Requirements can be specified before build.	Behavior must be experienced before it is trusted.
 What drives design		Known happy path drives the design.	Unknown edge cases drive the learning agenda.
 Role of launch		Launch confirms the roadmap.	Launch starts the real feedback cycle.
 Quality focus		Quality mostly means bugs and UX friction.	Quality means correctness, usefulness, safety, trust.

Discovery

How should teams discover AI use cases?

Traditional product development

Interview users and validate needs.

Study pain points and unmet needs.

Ask whether users want the feature.

Prototype after the opportunity is clear.

Domain knowledge helps prioritization.

AI-driven product development

Do that, then add domain experts to define “good”.

Observe manual workflows where humans inspect, decide, summarize.

Ask whether AI can reliably help in the workflow.

Prototype while the opportunity is still being learned.

Domain knowledge is required for evaluation and prompts.

Context is King

What does the model need to “see”?

Traditional product development

Context is mostly product state and user input.

Designers ask what screen to show.

More information rarely breaks simple features.

Success is predictable output.

Data access is implementation detail.

AI-driven product development

Context is the product: files, history, logs, permissions, intent.

Teams ask what context to retrieve and when.

Too much context can dilute relevance or exceed limits.

Success is the right information at the right moment.

Context strategy is core product strategy.

Testing

How do you know the AI product is good?

Traditional product development

Use beta testing and A/B tests.

Measure adoption, conversion, retention.

QA checks expected paths.

Feedback after release is useful.

One release stabilizes the feature.

AI-driven product development

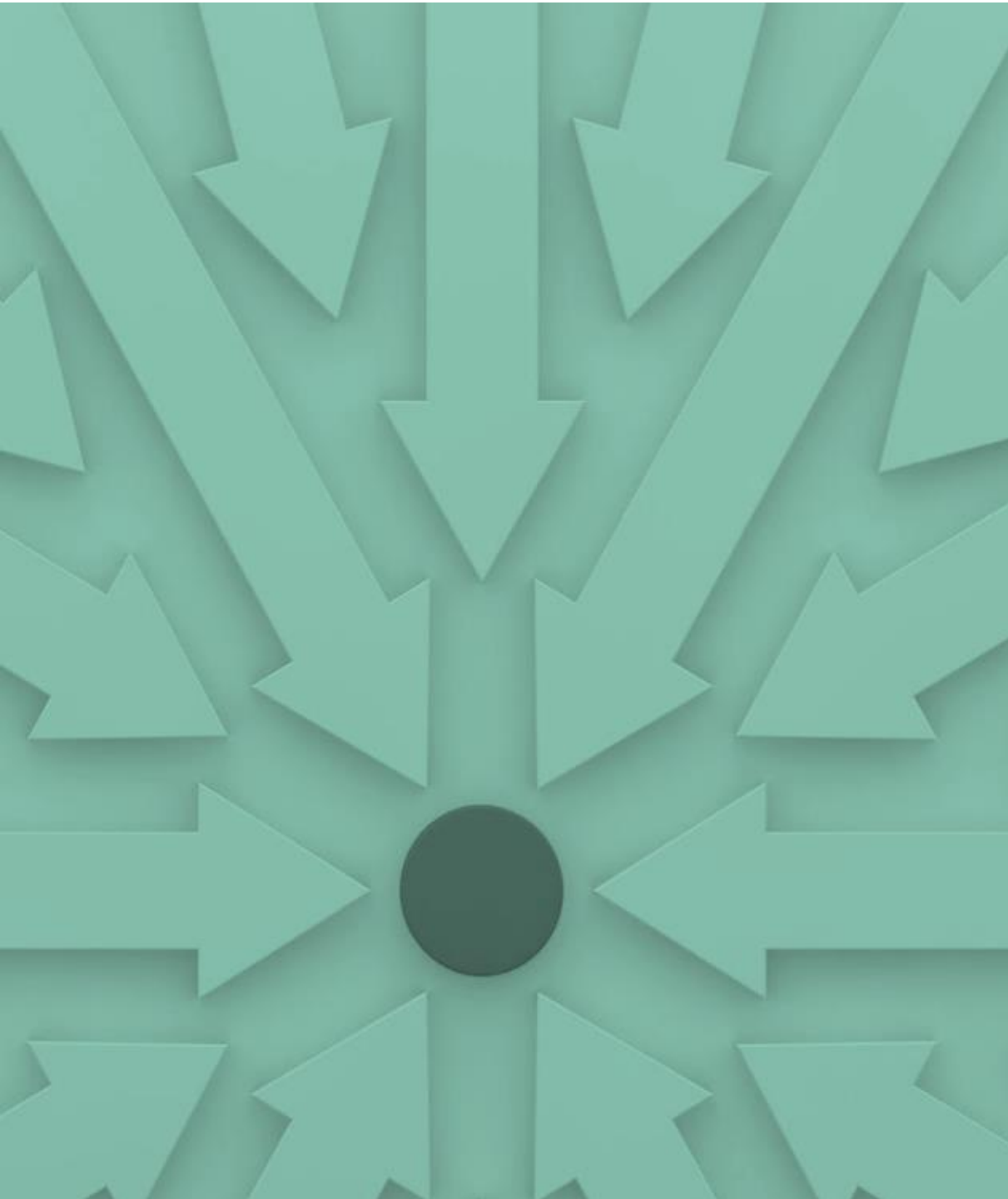
Add expert review, eval sets, and observed workflows.

Measure usefulness, correctness, trust, and saved effort.

Evaluation hunts long-tail edge cases.

Feedback loops are part of the product architecture.

Model/prompt/context changes keep the feature alive.



Use AI to **shorten
the loop** between
building and
learning.

Let's Talk about the Hype

What is hype?



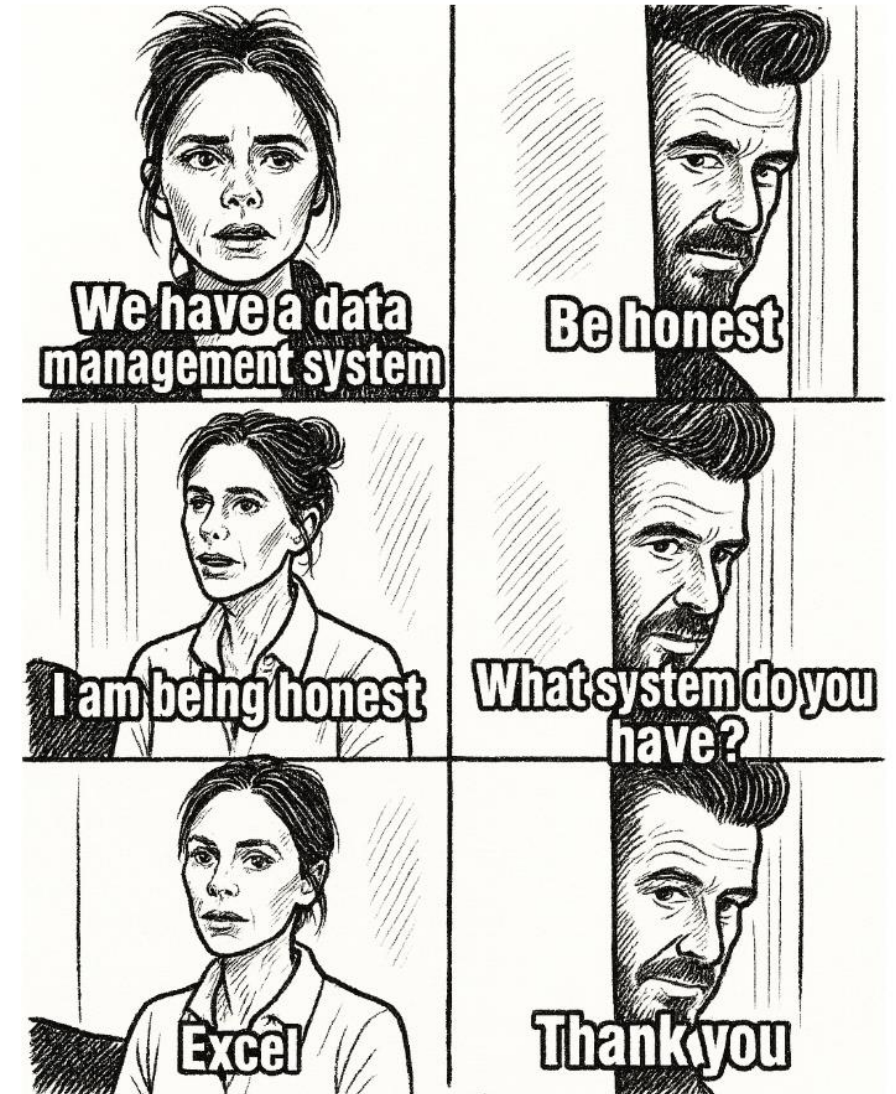
Urban Dictionary

<https://www.urbandictionary.com/define?term=hype>

hype

A fad. A clever marketing strategy which a product is advertized as the thing everyone must have, to the point where people begin to feel they need to consume ...

If you cannot manage hype, you will fail at your job

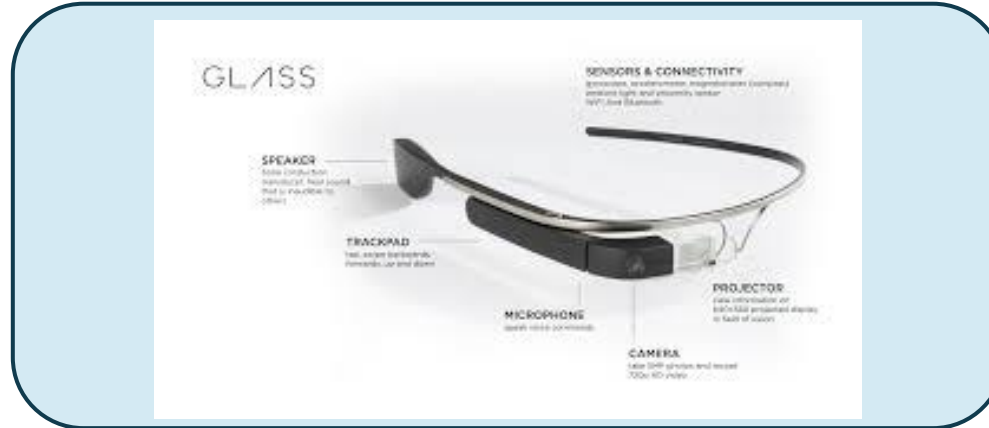


Hype

Apple AirPower



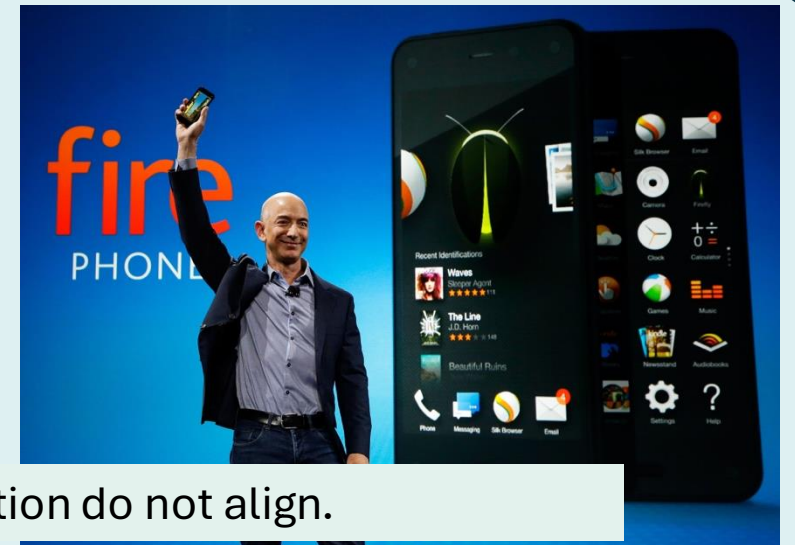
Google Glass consumer version



Apple Car / Project Titan



Amazon Fire Phone



Products fail when promise, timing, feasibility, and adoption do not align.

Why Products Fail

Reason for failure

Engineering was too hard

Scientific assumption failed

Regulation or political resistance

Missing ecosystem

Weak product-market fit

Social acceptance problem

Too expensive for the value offered

Internal company strategy conflict

Too early for the market/infrastructure

Physical-world complexity underestimated

Users did not trust long-term support

Example products

Apple AirPower; Google Project Ara; Magic Leap One

Google/Verily glucose-sensing smart contact lens

Facebook/Meta Libra/Diem; **Apple Car**

Amazon Fire Phone; Google Stadia; Microsoft Kin

Google Wave; **Google Glass**; Amazon Fire Phone;
Google Stadia

Google Glass; Facebook/Meta Libra/Diem

Google Glass; Microsoft Kin; **Amazon Fire Phone**

Microsoft Courier; **Apple Car**; Google Project Ara

Google Glass; Magic Leap One; Google Stadia

Apple Car; Google Project Ara; Magic Leap One

Google Stadia; Google Wave; **Google Glass**